



Design of a dual pH and temperature responsive hydrogel based on esterified cellulose nanocrystals for potential drug release

Hossam E. Emam^{*}, Tharwat I. Shaheen^{*}

Department of Pretreatment and Finishing of Cellulosic based Textiles, Institute of Textile Research and Technology, National Research Centre, Scopus affiliation ID 60014618, 33 EL Buhouth St., Dokki, Giza 12622, Egypt

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ABSTRACT

In this study, new stimuli – responsive hybrid hydrogels were achieved via succinylated cellulose nanocrystals (Su-CNC). The innovation was concerned with the inclusion of Su-CNC, at different degree of substitution (DS), into hydrogel network to render it pH and thermo-responsive characters through free radical polymerization reaction with poly(N-isopropylacrylamide) (PNIPAm). The prepared hydrogel was also examined for the in vitro release of Famotidine at different pH values. As clearly evident from the results, all the hydrogels prepared with different DS of Su-CNC, which were nominated as Su-CNC / PNIPAm (1–3), showed a high response to temperature change since their swelling behavior and hydrophilicity were decreased at 35 °C and upwards. This led to the more hydrophobicity character and thus the hydrogel shrinkage occurred. On the other hand, at pH 6, the hydrogels exhibited a significant Equilibrium Swelling Ratio (ESR) attaining 18.1, 17.3 and 16.8 (g/g) for Su-CNC / PNIPAm (1–3), respectively. However, Su-CNC / PNIPAm 2 hydrogel showed a significant response to the pH change from 8 to 2 which was advised to be selected as a potential pH responsive hydrogel for the in vitro Famotidine release.

1. Introduction

In the last few decades, great efforts from the researcher were devoted for developing of multiple responsive hydrogels, where changing in external triggers causes an intrinsic changing in their properties including swelling, porosity, mechanical and physical properties (Sun et al., 2020). Multi-responsive hydrogels are of tremendous interest because they contribute to intelligent scientific research including, drug delivery, tissue culture, bio-separators and bio-sensors. The external stimuli are commonly categorized to physical triggers e. g., temperature, light, magnetic and electrical fields, while the redox reaction, ionic strength and pH of the medium are examples of chemical triggers (Haq et al., 2017; Jiang et al., 2020; White et al., 2013). Owing to these unique properties of the hydrogel, it has opened enormous possibilities for emerging them in the diverse applications, as those composed of PNIPAm, making them a source of great interest in the field of drug delivery as a thermal responsive hydrogel (Sun et al., 2020).

Thermo-responsive hydrogels are found to have different behavior regarding their hydrophilic / hydrophobic polymer balance which recognized as law critical solution temperature (LCST). PNIPAm exhibited LCST at around 32 °C, near to human body temperature,

which can be displayed as a potent ideal material for biomedical applications. At temperature low LCST, all the components of hydrogel system are quietly miscible in all proportions and, therefore, it swells easily in aqueous solution. Above this LCST, the hydrophilic character decreases due to diminishing of the intermolecular hydrogen bond. On the other hand, the major property of the pH-responsive polymers is their eminent affinity to release/accept proton respecting the change of the ionic strength of medium (A. Hebeish et al., 2015; Hussain et al., 2021). Consequently, the polymer tends to swell at high pH values, because of the electrostatic repulsion of the negatively charged groups. Their sensitivity to pH changing can be motivated through employing a polymer composed of weak polyelectrolytes (i.e., polyacrylamide (PAAm), poly(*N,N*-dimethylaminoethyl methacrylate) or poly(–acrylic acid) (PAA)), which exhibits a critical phase shift from collapsed to swollen when ionized (Patil et al., 2021). Alternatively, natural polymers such as alginate, albumin, chitosan, etc., can also display pH responsive character.

Most recently, different pH responsive polymers have been incorporated into PNIPAm-based hydrogels for inducing dual pH/thermo-responsive hydrogels (Yue et al., 2020). The 3D interpenetrating polymeric network (IPN) hydrogels, are consisting of at least two crosslinked

^{*} Corresponding authors.

E-mail addresses: hossamelemam@yahoo.com (H.E. Emam), Shaheen_chem@yahoo.com (T.I. Shaheen).

natural and/or synthetic polymers, are capable for adapting their volume changes through swelling/deswelling in response to the multitude of external stimuli e.g., pH and temperature. This could be attributed to that each polymeric chain within IPN can manifest its own unique responsiveness to an environmental change (Liu et al., 2017). The salient features behind the use of stimuli-responsive hydrogels in terms of drug delivery are their tendency to release drugs that can be controlled at the intended target which, in turn, gains several advantages, notably lowering the therapeutic doses which reduce the dose side effects. Moreover, the incorporation of a PNIPAm IPN hydrogel with different polymeric moieties reveals an improvement in its physical properties compared to pure PNIPAm (Gupta et al., 2002).

Due to the growing awareness in biodegradable and renewable materials, biopolymers such as cellulose and its chemical modifications have recently gained more attention (Tharwat I Shaheen et al., 2021). Cellulose nanocrystal (CNC) has been harnessed to support IPN systems as reinforcement agent. Besides, surface modification of CNC has been also developed through etherification and esterification via using different approaches. Wan Ishak and coworkers reported the successful preparation of pH responsive PAA hydrogel reinforced CNC using gamma radiation (Wan Ishak et al., 2020). Several other polymers have been grafted onto CNC via free polymerization reaction to impart them pH responsive behavior. Of these polymers are, poly[2-(dimethylamino) ethyl methacrylate] (Tang et al., 2014), poly(4-vinylpyridine) and poly (N-[3-(dimethylamino)propyl]) methacrylamide (Kan et al., 2013). As far as we know, although the multi-responsive systems including hydrogels have been developed recently, there are a few reports concerned with the use of the grafted CNC in preparation of pH/thermo-responsive hydrogels.

In addition, our previous work has demonstrated the role of CNC-graft-PAAm in the preparation of multi-responsive pH/thermo-responsive hydrogel (A. Hebeish et al., 2015). However, incorporation of PAAm in IPN hydrogel resulted in a phase change in the alkaline medium that is not favorable for oral drug release because PAAm is fully hydrolyzed in an alkaline medium making it unstable (Gunari & Gundiah, 1981; Kurenkov et al., 2001). In the present work, we aimed to exploit the carboxylate anion groups of our previously prepared Su-CNC (Emam & Shaheen, 2019), a CNC esterification of with succinic anhydride, in the preparation of stable multi-responsive hydrogel with high amenability to release drug in acidic medium.

Precisely, three different series of Su-CNC (1–3), prepared using different DS ratios, were incorporated in free radical polymerization system with PNIPAm in order to obtain IPN multi-responsive hydrogels. We proposed that, COO[−] groups from Su-CNC can, effectively, impart a new pH character to the IPN of the thermal responsive PNIPAm hydrogel. This work also has addressed the optimization of the potential properties for using the temperature- and pH-responsive Su-CNC / PNIPAm hydrogel through investigation the impact of DS ratios of Su-CNC on the responsive behaviors of the formed hydrogels. Releasing from the optimal hydrogel was evaluated via the kinetic study to examine its advantageous as a template for drug release, wherein Famotidine was chosen as a drug model. In common practice, Famotidine is an active material used in the stomach pain relievers through diminishing of the production of the stomach acid juice and principally used to treat peptic diseases. Physicochemical characterizations were conducted as well as the drug release kinetic study was also performed.

2. Experimental

2.1. Materials

Succinic anhydride (≥ 99%), N-isopropyl acrylamide (NIPAm, 99%), N,N'- bismethylene acrylamide (BMA, 99%), N,N-Dimethylacetamide (DMAC, ≥99.5%) and potassium thiosulphate (K₂S₂O₈, 99%), were all supported from Sigma Aldrich (Germany) and were used as received without purification.

2.1.1. Preparation of temperature and pH responsive hydrogels

2.1.1.1. Preparation of Su-CNC. CNC was synthesized by using the previously reported methods (El-Naggar et al., 2018; Hebeish, Farag, Sharaf, Shaheen, 2014b; A Hebeish et al., 2018; Th I Shaheen & Emam, 2018; Th I Shaheen et al., 2019). Su-CNC was fabricated by esterification reaction of CNC with succinic anhydride as mentioned in our previous work (Emam & Shaheen, 2019) in three different DS ratios. In brief, three different weight ratios CNC: succinic anhydride (1:5, 1:9 and 1:12) were mixed together in round flask under continuous stirring at 115 °C for 6 h and labeled as 1, 2 and 3, respectively. Afterwards, DMAC (50 mL) was used for avoiding the recrystallization of the free anhydride through adding to the mixture. The produced Su-CNC was neutralized prior drying at 60 °C. The typical DS of the synthesized Su-CNC, at different feeding ratios, was estimated via assigning their carboxyl content.

2.1.2. Preparation of the responsive hybrid hydrogels

The responsive Su-CNC/PNIPAm hybrid hydrogel was prepared by the free radical polymerization of Su-CNC with the NIPAm monomer. Both concentrations of reactants; Su-CNC/NIPAm, were kept at 90:10 (v/w). In short, Su-CNC was re-suspended in deionized water before addition of monomer. Then, the monomer of NIPAm and crosslinking agent BMA were added to the suspension solution for 30 min under vigorous stirring. Initiator (K₂S₂O₈; 1.5% of monomer weight) was then dissolved and drop-wisely added to the latter suspension. Polymerization reaction was started after the temperature attained 70 °C for 1 h. At the end of reaction, the as-formed hydrogel was purified by immersing in deionized water and rinsing several times for removal of impurities prior drying in freeze-drier (−60 °C, 0.1 mbar). For evaluating of the effect of DS of Su-CNC on the responsiveness of the prepared hydrogel, the synthesized Su-CNC with different DS was used to prepare Su-CNC/PNIPAm (1-3) hydrogels. The graft yield % (gel fraction content) was calculated according to the following Eq. (1),

$$\text{Graft yield\%} = (\text{Wh} - \text{Ws})/\text{Ws} \times 100 \quad (1)$$

Where Wh and Ws were the weights of the hydrogel after freeze-drying and substrates, respectively.

2.2. Characterizations

2.2.1. Determination of DS of Su-CNC

The DS of samples were calculated by estimating their carboxyl content according to following equation (eq. 2) (Zhong et al., 2012). Initially, the contents of carboxylic (COOH) groups in the synthesized samples were estimated by the methylene blue (MB) solution method with a fixed concentration 0.00313 μmole/mL according to reports in literature (Emam et al., 2017; Emam & El-Bisi, 2014).

$$\text{DS} = (162 \times n_{\text{COOH}})/(m - 100 \times n_{\text{COOH}}) \quad (2)$$

Where $n_{\text{COOH}} = [(C1 - C2) \times 0.00313/m]$, is the molar no. of COOH group (mmol/kg), C1 & C2 are MB concentration (mg/L) for blank & samples and m is the weight of sample (g). Also, 162 and 100 refer to the molar mass of AGU and the net increase of the mass of an AGU for each succinic substituted, respectively.

2.2.2. Physicochemical characterizations

Characterization of Su-CNC and Su-CNC/PNIPAm hybrid hydrogels under investigation was performed using the world-class facilities. Chemical structure changes were monitored using Fourier transform infrared spectroscopy (ATR - FTIR, Jasco FT/IR 6100). The measurements were performed by attenuated total reflection (ATR) mode (Cha et al., 2012).

The porous structure of the prepared hydrogels was examined morphologically by scanning electron microscope (SEM, JSM-6400,

JEOL–Japan). In addition, the thermal properties of hybrid hydrogels prepared at different DS were investigated using TGA/SDT Q600 analyzer instrument.

2.2.3. pH/thermo responsive character of the hybrid hydrogels

Responsive characters of the prepared hydrogels was evaluated regarding to their swelling behavior at different temperatures and pH media, which measured by the classical gravimetric method and expressed as ESR. Therefore, their swell ability was also measured at different pH (2–10) and different temperature (20–60 °C). The equilibrium absorbed water was calculated using Eq. (3),

$$ESR = \frac{W_1 - W_0}{W_0} \times 100 \quad (3)$$

where W_0 is the sample weight before and W_1 is the weight after swelling, respectively (Cha et al., 2012; Ekici, 2011; Hebeish et al., 2014c).

2.2.4. Drug release study

Famotidine as model of drug was loaded independently on the Su-CNC/PNIPAm hydride hydrogel. 100 mg of freeze-dried hydrogel was firstly incubated and then immersed in concentrated drug solution (2

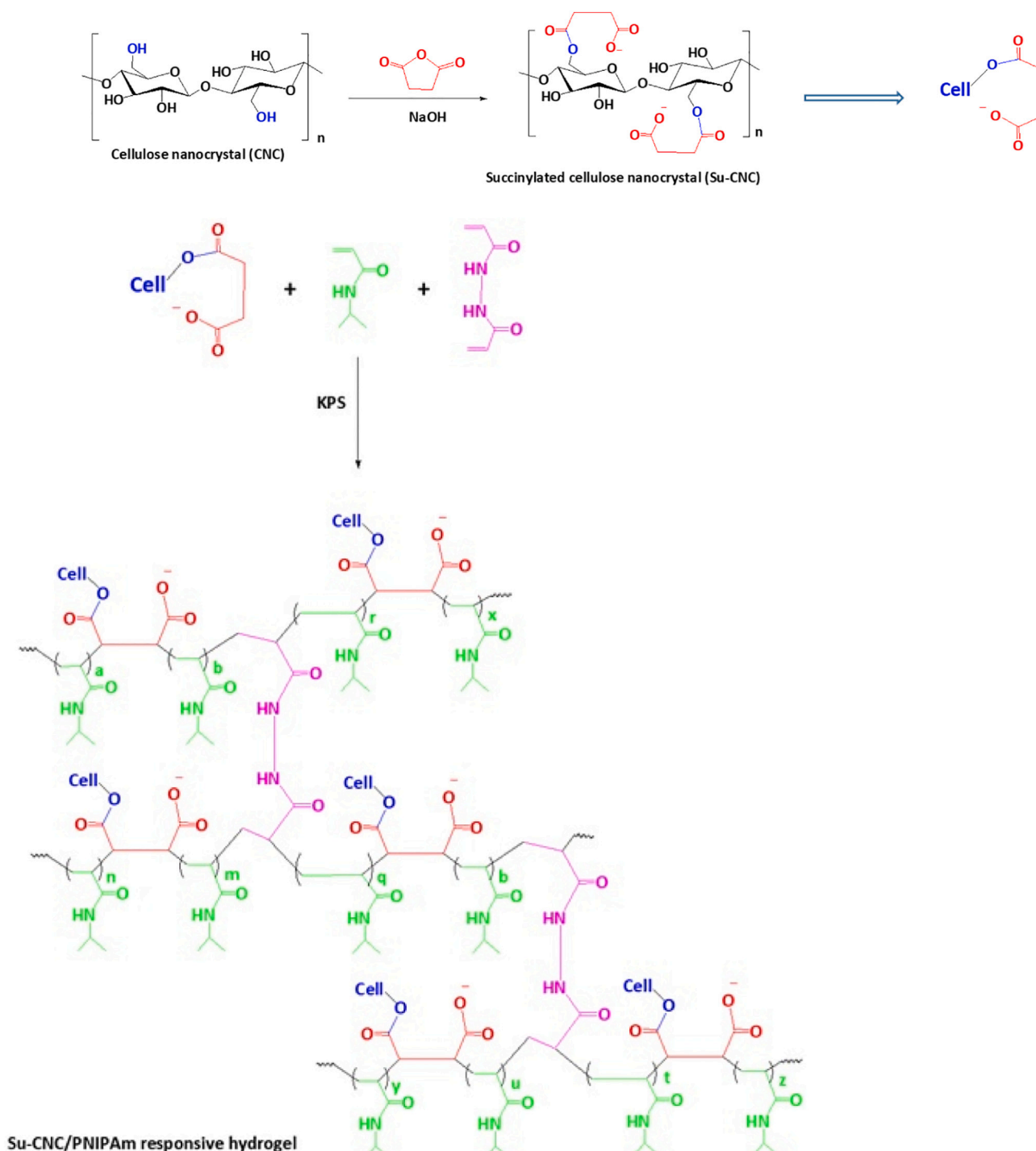


Fig. 1. Schematic mechanism for preparation of Su-CNC/PNIPAm hybrid hydrogel.

mg/2 mL) for 2 h at 25 °C (Petrusic et al., 2012; Salehi et al., 2009). Afterwards, the swollen drug loaded samples were decanted from the solution and dried at 70 °C until constant weight prior to drug release studies.

In-vitro study of Famotidine release from the loaded Su-CNC/PNIPAm hydrogel was carried out in different medium (acidic & alkaline) at 37 °C. The drug loaded Su-CNC/PNIPAm hydrogel was placed in a dialysis membrane (purchased from Sigma Aldrich; cut off: 2000) and then floated in a flask containing 20 mL of solution (pH = 2 & pH = 10). The released drug of Famotidine was spectrophotometric estimated by using Cary-100 UV-Vis-NIR Systems at the characterized wavelength of 276 nm. The results of cumulative release were shown as a function of time (0–360 min), in which 2 mL sample was withdrawn from the mixture for absorption measurement. The amount of Famotidine released from hydrogel to the solution was estimated from the absorbance intensity normalized to the calibration curve. The release was performed in replicates. To understand the release mechanism, kinetic of Famotidine release was systematically studied by applying four different kinetic models represented in Higuchi, Hixson-Crowell and Korsmeyer-Peppas.

3. Results and discussion

3.1. Preparation of Su-CNC/PNIPAm hybrid hydrogel

The hybrid hydrogel was synthesized based on CNC in two separated steps; namely esterification and grafting as presented in Fig. 1. The esterified CNC (Su-CNC) was produced by the interaction of succinic anhydride with CNC. The anhydride was bonded with the cellulose through the condensation reaction with the alcoholic groups of cellulose forming ester. The condensation was supposed to occur at the primary alcoholic groups in the C6, followed by secondary alcoholic groups in the C2 and C3 (Xin et al., 2017). The DS of the as-synthesized samples at different feeding ratios of succinic anhydride were 1.4, 1.8 and 2.1 for 1, 2 and 3, respectively.

In the next step, the IPN of hybrid hydrogels was obtained through free radical polymerization of PNIPAm with Su-CNC in the presence of BMA (Fig. 1) and KPS. Initiation reaction was started by abstracting hydrogen from C—H bond of Su-CNC, where the free radicals were generated. These radicals attacked the double bonds (—C=C—) of NIPAM and/or MBA leading to form further free radicals which were reacted with each other to form IPN structure of /PNIPAm hydrogel. Furthermore, by increasing DS of Su-CNC, it is plausible that, the crosslinking accessibility was enhanced with an increment in the number of freely accessible succinate groups. The aforementioned suggestions reflected in generation of highly porous IPN of the as-required Su-CNC/PNIPAm hybrid hydrogel. The graft yield % was also varied regarding to the DS used. The values of graft yield were $57.5 \pm 1.6\%$, $68.7 \pm 2.3\%$ and $60.78 \pm 1.2\%$ for Su-CNC/PNIPAm (1–3), respectively. These variations in graft yield % are significantly related to DS of Su-CNC influence on the content of gel fraction among the prepared hydrogels. The dramatic increment of the gel fraction with the DS of Su-CNC was due to increasing of the active centers during initiation step, where, the grafting polymerization occurred. Further increase in DS, the steric hindrance effect restricted the formation of active centers as well as shortened the length of PNIPAm chains, resulting in lower gel fraction than expected.

The FTIR spectral analysis for CNC, Su-CNC and Su-CNC/PNIPAm hydrogel were measured as shown in Fig. 2 to detect the interchanging in their chemical structures corresponding to the polymerization process. The CNC exhibited five spectral absorption bands, characterized for the cellulose macromolecule, at 3292 – 3324 cm^{-1} , 2892 cm^{-1} , 1624 cm^{-1} , 1417 cm^{-1} and 1010 cm^{-1} (Hebeish et al., 2013; Hebeish, Farag, Sharaf, Shaheen, 2014a). These absorption bands were attributed to the stretching vibration of O—H, asymmetric stretching of aliphatic C—H, stretching of C=O, the scissoring of H-C-H

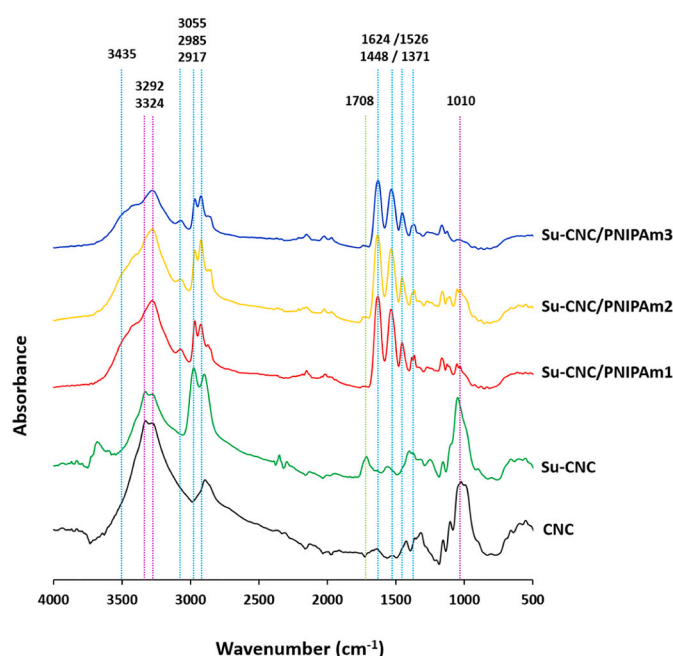


Fig. 2. The infrared spectra for the prepared Su-CNC/PNIPAm hybrid hydrogels in comparison with CNC and esterified CNC.

and O—H bending, respectively. After the esterification, three new absorption bands were obviously appeared at 3670 cm^{-1} , $2977/2886$ cm^{-1} and 1708 cm^{-1} which were assigned to the succinic acid. These bands were attributing to the O—H stretching, stretching of aliphatic methylene and carbonyl of the ester group formed between anhydride and cellulose, respectively (Emam & Shaheen, 2019). Moreover, the characteristic bands of cellulose became less intense relating to the esterification process. These spectral data declared the esterification reaction of cellulose and further confirmed the covalent bonding between cellulose macromolecules and succinic anhydride forming ester of Su-CNC.

After the grafting process, the Su-CNC/PNIPAm hybrid composite showed several new absorption bands at 3435 , 3055 cm^{-1} , $2985/2917$ cm^{-1} , 1624 cm^{-1} , 1526 cm^{-1} , 1448 cm^{-1} , 1371 cm^{-1} and 1103 cm^{-1} . The highlighted new bands were specified for the free NH_2 , crosslinked NH_2 , CH_3 , C=O, NH_2 - (amide), NH - (amide), N-(amide) and C—C bending, respectively (A Hebeish, Farag, Sharaf, Shaheen, 2014b; Wei et al., 2016; Zhu et al., 2012). Additionally, the absorption bands of the ester linkage between CNC and anhydride (1708 cm^{-1}) and the O—H bending were both significantly diminished attributing to the grafting. These spectral analyses referred obviously to the grafted PNIPAm with Su-CNC, where the BMA acted as crosslinker.

The morphological features of the synthesized Su-CNC/PNIPAm hybrid hydrogel were studied under the electron microscope via examination of the cross-section for the swelled hydrogels as seen in Fig. 3. The synthesized hybrid hydrogels manifested a homogeneous network structure that is described with irregular, well-distributed pores. These pores were interconnected with each other in a crosslinking network, that abled for easy flow the entrapped drug molecules outside. Indeed, the pore size of the hydrogels under investigation was measured using the 4 pi analyses software, which appeared closer to each other in the range of 5 – 20 μm , which was a consistent with the previous reported works (Evans et al., 2021; Whang et al., 1999). It is noticeable that, the size of pores was slightly increased by increasing the accessibility of cross-linking, which achieved by increasing the DS of Su-CNC used in the hydrogel network formation. As per the schematic mechanism of hydrogel 3D network formation (Fig. 1), the succinate group of Su-CNC involved in the free chain polymerization reaction was allowed to

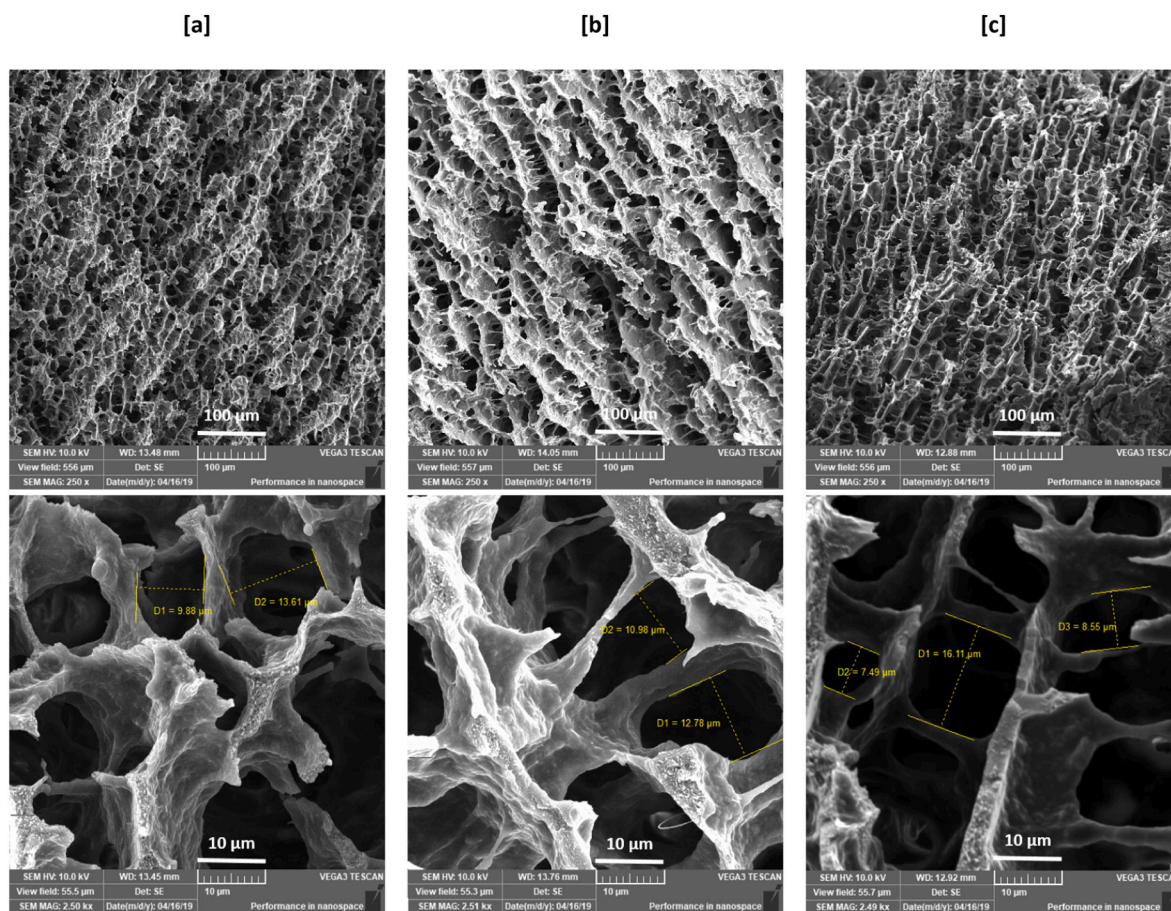


Fig. 3. The microscopic images for Su-CNC/PNIPAm hybrid hydrogels at two different magnifications (250 $\times$ and 2500 $\times$); [a] Su-CNC/PNIPAm1, [b] Su-CNC/PNIPAm2 and [c] Su-CNC/PNIPAm3.

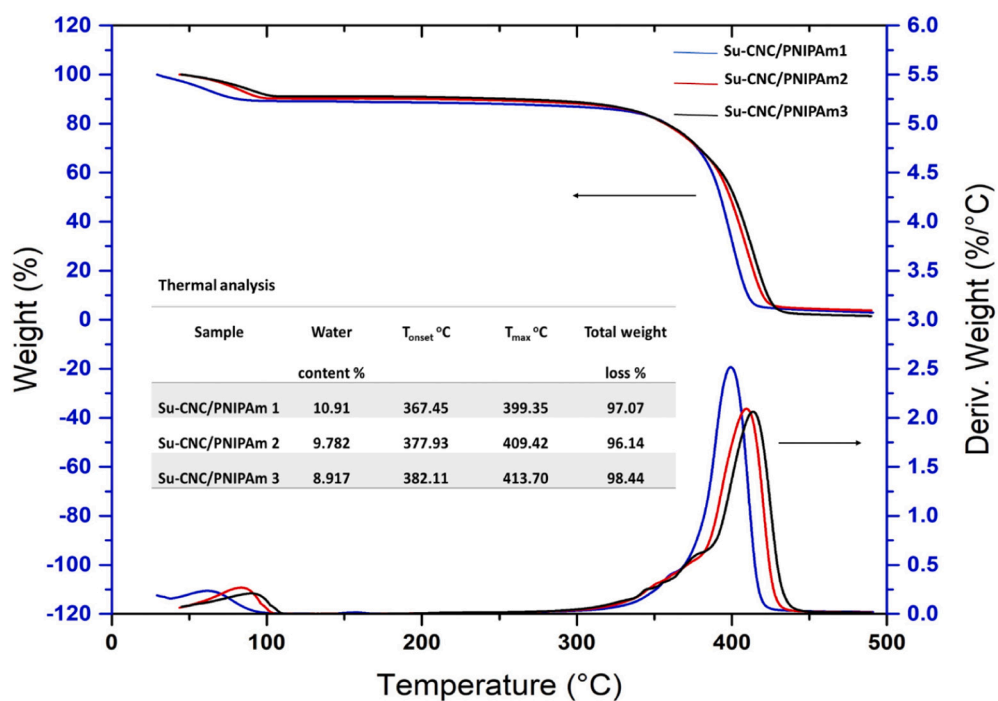


Fig. 4. TGA (left scale), DTGA (right scale) and Thermal analysis (disclosed from figure) of Su-CNC/PNIPAm hybrid hydrogels.

generate two active centers which are accessible to initiate different two chains. By increasing DS, more succinate groups and therefore more active centers were readily allowed to initiate the polymerization reaction and, thus, chains with crosslinking potentials were increased. Consequently, at highest DS, the as-formed hydrogels manifested thick-bridges which, in turn, limited their swelling capacity. These microscopic observations revealed a significant influence on the swelling properties of the hydrogels.

3.2. Thermal properties of Su-CNC/PNIPAm hybrid hydrogel

Thermographic behavior (TGA) of the prepared Su-CNC/PNIPAm hybrid hydrogels and their first derivatives (DTGA) at different DS were plotted in Fig. 4 accompanying with their decomposition values occurred at the active pyrolysis region (260–450 °C) through comparing their onset temperature (T_{onset}) and maximum temperature (T_{max}).

It can be seen from TGA data that, the moisture/water contents, which attributed to the grafted PNIPAm of hydrogels were varied according to the DS of Su-CNC. Since, the more hydrophilic surface character is the more grafted PNIPAm content. Further, the variation in DS of Su-CNC/PNIPAm hybrid hydrogels slightly enhanced the onset temperature during thermal degradation which might reflect the increment of thermal stability increased by increasing the ratio of Succinic groups. Nevertheless, the thermal profile of Su-CNC / PNIPAm2 hydrogel manifested a close degradation behavior to Su-CNC / PNIPAm 3, it has a lowest weight loss % than other. These findings precisely indicate that, Su-CNC / PNIPAm 2 hydrogel can be nominated as a most prominent candidate for drug release based on their thermal decomposition stability profile.

3.3. The pH-thermo responsive and swelling properties of Su-CNC/PNIPAm hydrogel

The responsive ability of hydrogels was evaluated as a function of their volume phase change. Noticeably, the swelling profile was known to be a key feature for understanding the release behavior, and therefore, the swelling properties of the prepared hydrogel was an important issue to be studied. In this regard, the effect of different DS of Su-CNC on the ESR of the prepared Su-CNC/PNIPAm hybrid hydrogels was studied at different contact time and pH ranging from 2 to 10 as clearly represented in Fig. 5. Results clearly declared that, the ESR of hydrogels was increased gradually with increasing the contact time that they collapsed in solution. Further, increasing the DS of Su-CNC from 1.4 to 2.1 was accompanied by decrement in the swelling ratio over all the investigated hydrogels. Regardless of gel fraction content %, the higher DS of Su-CNC led to increase crosslinking between Su-CNC and PNIPAm during formation of IPN structure of hydrogel. However, in the presence of some external triggers such as pH and temperature, the swelling behavior of the hydrogel was altered in response to external stimuli based on the chemical groups attached to the polymeric chains.

As far as was known, succinic acid owned two different pKa occurring at 4.21 and 5.41, which after esterification with CNC, the remained free carboxyl group could dissociate at pKa 5.41. Therefore, all the hybrid Su-CNC/PNIPAm hydrogels showed a maximum swelling ratio at pH > pKa (pH 6). The swelling affinity of hydrogel was principally influenced by the protonation and ionization balance of the carboxyl groups. At low pH (< pKa), the carboxylic groups from Su-CNC in hydrogel were prominently protonated which, in turns, restricting the electrostatic repulsion force, arisen because of ionized carboxylic groups ($-\text{COO}^-$), that induced the intermolecular hydrogen bond between carboxylic groups and thus, the polymer – polymer interaction prevailed instead of polymer-water interaction. This certainly led the network of hydrogel to shrink and consequently the swelling of hydrogel was significantly decreased. Conversely, the pH as high pH 8 (> pKa), most of carboxylic groups were ionized, leading to dissociation of intermolecular hydrogen bonds. As a result, the electrostatic repulsion among

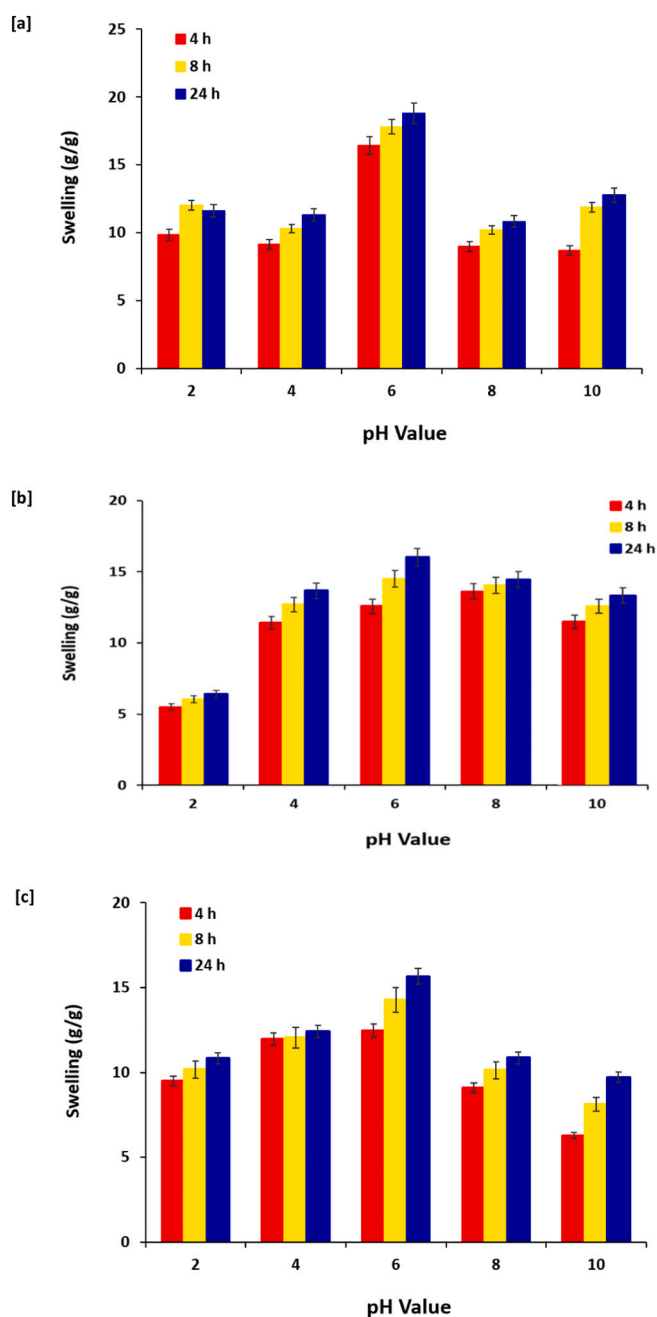


Fig. 5. The swelling of the prepared Su-CNC/PNIPAm hybrid hydrogels as function of pH and time; [a] Su-CNC/PNIPAm 1, [b] Su-CNC/PNIPAm 2 and [c] Su-CNC/PNIPAm 3.

carboxylic groups ($-\text{COO}^-$) was arisen, which was useful in the relaxation of network to reach maximum ESR. Fast relaxation of the hydrogel network allowed the distance between different chains to be enlarged, which is evidently influenced by the variation of the DS of Su-CNC in the IPN of hydrogel as seen from Figs. 5 (a - c). The most potent stimuli responsive hydrogel was observed in case of Su-CNC/PNIPAm 2 of DS = 1.8. These could be due the gel fraction content % of the latter hybrid hydrogel was higher than others, where the rich cross-linked polymeric PNIPAm chains were existed. Nevertheless, at pH 6 (near neutral), all the hydrogel showed the high responsive ability toward swelling which might be due to the low counter ion concentration in the solution, resulting in the enhancement of the driven force for diffusing the water inside the IPN of hydrogel, giving rise to increase the swelling ratio at this pH value.

Therefore, the swelling ratios reached at pH 6 greatest values of 16.4, 12.6 and 12.5 g/g after 4 h to 24.8, 22.2 and 21.6 g/g after 24 h for /PNIPAm1 /PNIPAm2 and /PNIPAm3 hydrogel, respectively. Moreover, the increase of counter ions $[Na]^+$, pH from 8 to 10, may shield the negative charges of $-COO^-$ groups which weakening their electrostatic repulsion, bringing about a slight dropping in the swelling ratio varied as per DS (Hu et al., 2015).

On the other hand, the swelling behaviors of the hybrid Su-CNC/PNIPAm hydrogels were also studied at different temperatures in the range of 15–60 °C (Fig. 5), after reaching high ESR % at pH 6.5. Regardless to the DS of hydrogel, the swelling amounts of the hydrogel were insignificantly changed in the temperature less than 30 °C. While, the swelling amounts were sharply decreased from 21.6–17.8 g/g to 5.0–4.6 g/g by increasing the temperature from 30 °C to 35 °C and remained practically constant above this temperature. Furthermore, the swelling ability of hydrogel was at the maximum value displaying highest hydrophilicity at temperature lower than 35 °C. Otherwise, the hydrogel acquired a lowest swelling ratio showing hydrophobicity above 35 °C which was harmonious with the original LCST of PNIPAm. These data showed the sensitivity of the prepared hybrid hydrogel against pH and temperature of the contact medium which reflected in the control of releasing properties by pH and temperature. This means that the synthesized Su-CNC/PNIPAm hybrid hydrogel can effectively release the absorbed bioactive materials in either acidic pH or at temperature above 35 °C, which the acidic pH responsiveness was very useful in case of oral drugs. The mechanism for releasing of the active substance (drug) through pH and thermo-responsive IPN hydrogel in view of the previous assumptions was clearly pointed out in the schematic diagram as shown in Fig. 7.

The diagram showed the effect of pH on the polymer-polymer interaction induced by virtue of intra-molecular hydrogen bonds between the carboxylic groups of Su-CNC inside the hydrogel that occurred at low pH ($< pKa$), resulting in the release of active substances such as famotidine outside the pores. Moreover, the thermo-responsive as represented in the diagram through hydrogel shrinkage upon raising temperature up to LCST (>35 °C). Active materials or molecules could be varied respecting the application used. In this study, the target was the release of Famotidine which was usually taken orally due to its medicinal uses as reducing the amount of acid in the stomach. pH responsive character of the formulated hydrogel might allow control release of hydrogel according to the amount of acid presented in the stomach.

Thus, the pH responsive behavior of Su-CNC/PNIPAm 2 was examined at pH 2 and 10 by changing the pH as shown in Fig. 6, due to its high response to pH trigger with highest when compared with its ESR% in acid /alkaline media and others. Based on TGA and DTGA results, the

ash content was higher in the case of Su-CNC / PNIPAm 2 which revealed higher yields than others.

3.4. Release of Famotidine

Famotidine helped to reduce the production of stomach acid juice and principally used to treat gastro-esophageal, reflux peptic disease and ulcer disease. Therefore, the controlling of Famotidine release was quite necessary to organize the extreme benefit. Consequently, the release of Famotidine was examined in different two media; acidic (pH 2) and alkaline (pH 10). Primarily, Famotidine as gastrotidine injection was uploaded onto the synthesized /PNIPAm2 hydrogel and the loaded amount was nearly 60 mg/g. The release profiles of Famotidine were presented in Fig. 8 and the release was estimated ration (%) and amount (mg/g). Famotidine was gradually released by the contact time and the release was quite rapid in acidic medium (pH 2) as all Famotidine was almost released within 2 h only. In alkaline medium (pH 10), the applied hydrogel exhibited quite slow releasing rate and 85.3% of Famotidine (51.2 mg/g) was leached out from hydrogel after 6 h. For full release of Famotidine, 2 h more were needed. Compared to the releasing in pH 2, the release of Famotidine was controlled by increasing the pH to more alkaline. This hypothesis might be explained by the swelling behavior of Su-CNC/PNIPAm 2 hydrogel. The higher swelling behavior of the applied hydrogel was observed in alkaline pH and subsequently the Famotidine was retained inside the swelled structure of hydrogel and hence slowing down of the release was achieved. In contrast at acidic pH, the hydrogel was collapsed and most of Famotidine could be leached out easily.

3.5. Kinetic and mechanism of Famotidine release

Famotidine release from the applied Su-CNC/PNIPAm hydrogel was kinetically studied for understanding the releasing mechanism. Herein, Higuchi, Korsmeyer-Peppas and Hixson-Crowell as different kinetic models were studied for the Famotidine release as displayed in Fig. 9. For kinetic model of Higuchi, the release amount of Famotidine (mg/g) was plotted with the square root for time of release (SQRT) (Fig. 9a). While, the relationship between the cubic roots of the Famotidine release amount (mg) and time of release was described for the Hixson-Crowell kinetic model (Fig. 9b). For recognizing the Korsmeyer-Peppas kinetic model, the logarithm of Famotidine release amount (mg) was plotted with the time of release (Fig. 9c). The whole parameters of all kinetic models including R^2 , k_H , k_{HC} , n & k_{kp} were estimated and summarized in Table 1.

Based on Korsmeyer-Peppas data (Fig. 9c) and considering with the linearity of release profile in Fig. 8 (zero-ordered), the Famotidine released from the Su-CNC/PNIPAm hydrogel was complained with zero-ordered kinetic modelling. The principle of zero-order kinetic notified that the release behavior was basically independent on the Famotidine concentration uploaded on the hydrogel, however, it was significantly depended on pH of medium. Compared to acidic pH of 2, considerable slower Famotidine release was observed in alkaline medium (pH 10), which could be attributed to, more opportunity for formation of hydrogen bonding resulting in drug persistence within the hydrogel matrix (Sahin & Arslan, 2008). Data of Famotidine release showed best linearity ($R^2 = 0.99$) to Korsmeyer-Peppas kinetic and n values were 0.01 (≤ 0.5) which suggested the release of Famotidine may be performed via diffusion. Fig. 8a & b showed non-linear trend for both of Higuchi and Hixson-Crowell modellings. Hence, Famotidine release was oppositely proportional with surface area of hydrogel. Hydrogel characterized by high surface area in alkaline owing to the swelling behavior, while slow releasing was observed.

Based on the formerly reports in literature (Gull et al., 2020; Korsmeyer et al., 1983; Liakos et al., 2013; Papadopoulou et al., 2018; Su et al., 2020; Tong et al., 2020; Zhang et al., 2021), releasing of any loaded materials from polymeric skeleton could be physically taken

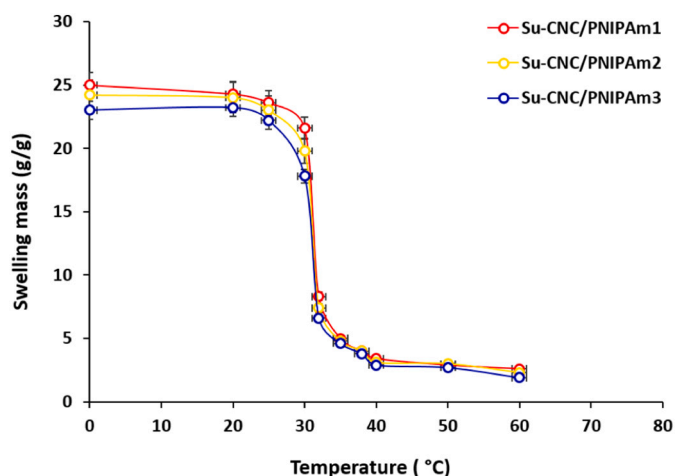


Fig. 6. The swelling profile of Su-CNC/PNIPAm hybrid hydrogels as function of temperature (thermo-responsive).

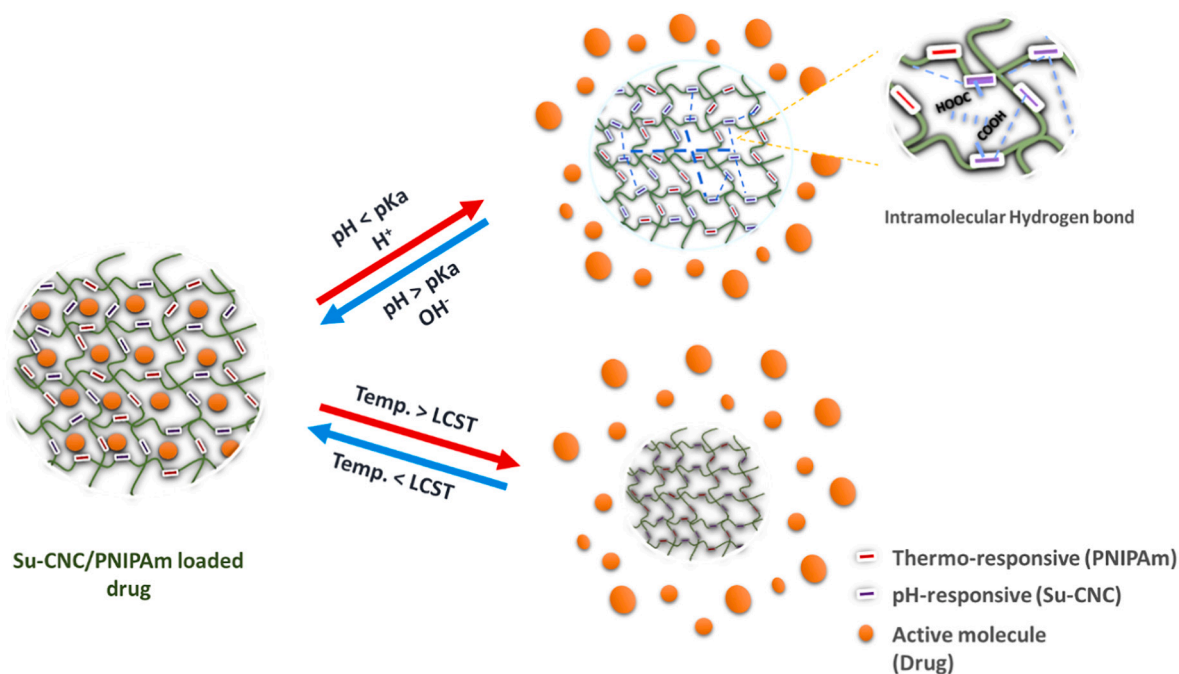


Fig. 7. Schematic diagram illustrating the effect of pH and thermo-responsive on the active material (drug) release.

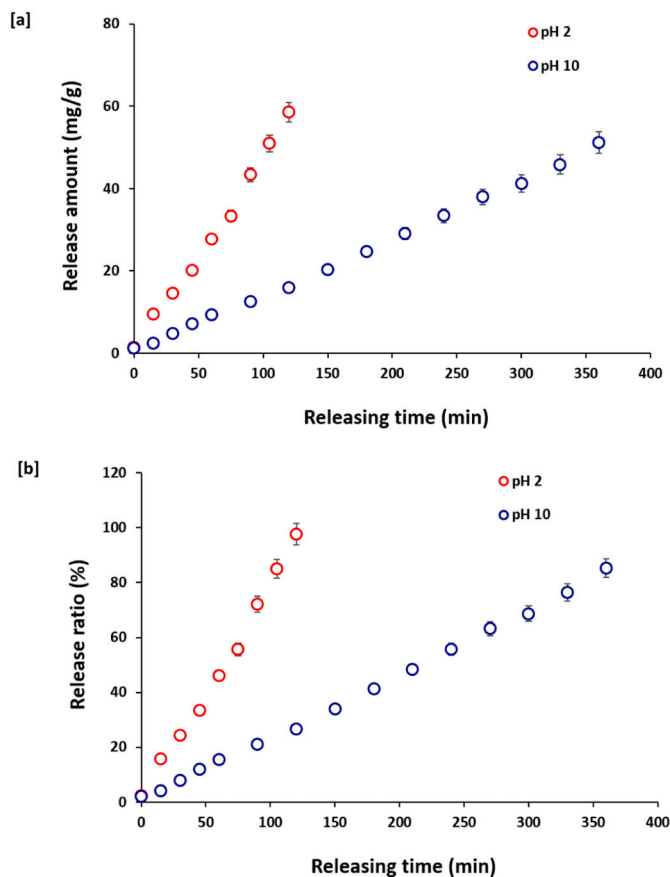


Fig. 8. Release profile of Famotidine from the prepared Su-CNC / PNIPAm 2 hydrogel as function of time; [a] release amount and [b] release ratio.

place by one mechanism of material diffusion, polymer degradation/dissolution or polymer swelling. According to all of the above-illustrated results, controllable releasing of the drug was mainly dependent on the

medium pH which was reflected in the hydrogel swelling. Consequently, the releasing mechanism of Famotidine might be elucidated by hydrogel swelling. However, Krosmeier-Peppas kinetic supported the diffusion mechanism. Therefore, the releasing might be explained by both mechanisms of swelling of hydrogel and diffusion of Famotidine within the hydrogel.

The accessible alcoholic groups in the Su-CNC/PNIPAm hydrogel might be weakly bonded with the functional groups of Famotidine (O, S, N) through H-bonds (Abdelhameed & Emam, 2019; Emam et al., 2019). When submerged in the alkaline pH, the Su-CNC/PNIPAm hydrogel was swelled, the internal networked structure was opened and the Famotidine was slowly leached out into the environment, achieving controllable releasing process. So, it could be summarized that, the release behavior of Famotidine was in agreement with and explained by the swelling characters of Su-CNC/PNIPAm hydrogel in addition to diffusion of drug under the effect of pH. Due to the collapsing of hydrogel in acidic medium, all Famotidine was liberated from inside of hydrogel and easily leached out resulted in fast release behavior. While in alkaline medium, the hydrogel was swelled and keep the Famotidine molecules stably persisted within the hydrogel matrix and consequently the release of Famotidine became controllably slow.

Recently, different polysaccharides (curdian, dextran and pullulan) in combination with dopamine were applied as biocompatible hydrogels in control release of different antibiotic and drugs (Su et al., 2020; Tong et al., 2020; Zhang et al., 2021). Release of povidone-iodine from hydrogels-based starch, chitosan and alginate were studied (Gull et al., 2020; Liakos et al., 2013; Papadopoulou et al., 2018). Meanwhile, no works were formerly studied the Famotidine release from biopolymers-based hydrogels. Release of drugs from chitosan and alginate-based hydrogels was carried out within less than 4 h (Gull et al., 2020; Liakos et al., 2013). Thus, the release of drugs from starch containing bio-elastomers was quite slow and reached 7 days (Papadopoulou et al., 2018). The releasing of Famotidine from Su-CNC/PNIPAm hydrogel herein was (≈ 8 h) quite slower than that of drugs from chitosan and alginate-based hydrogels, and was much plausible compared to the bio-elastomer.

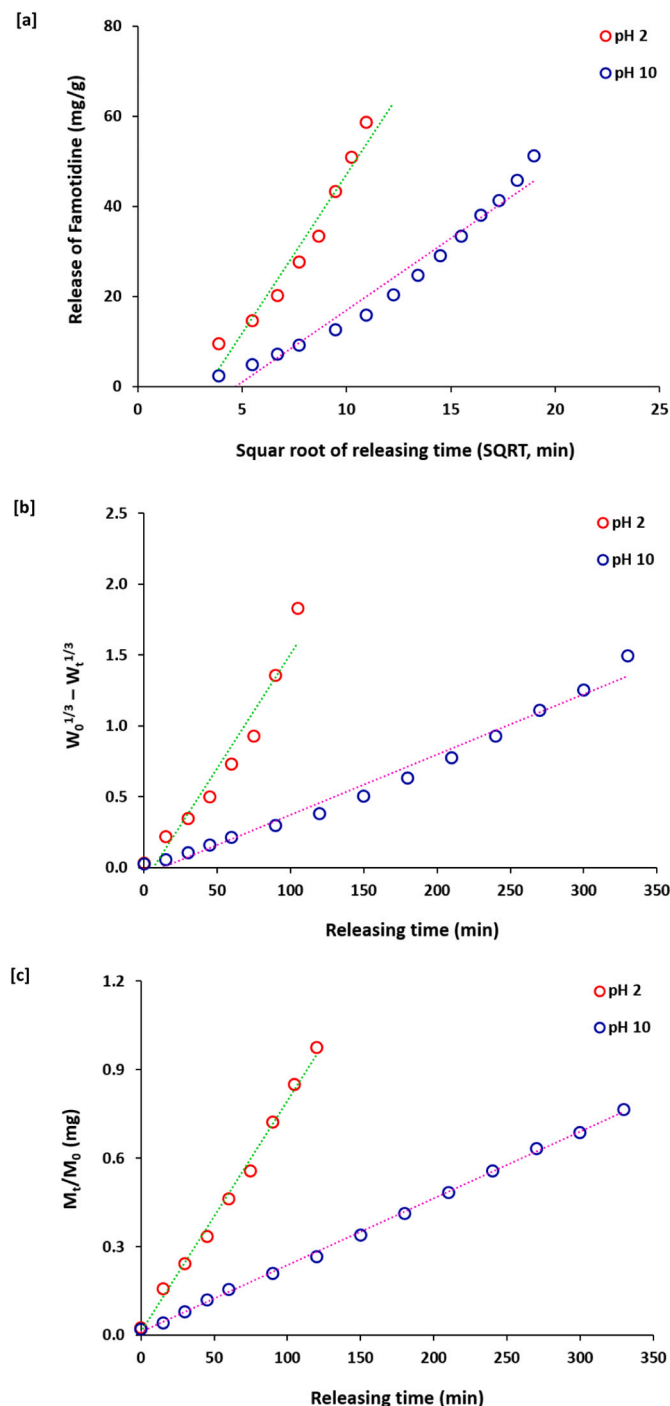


Fig. 9. Kinetics of Famotidine release from the prepared Su-CNC/PNIPAm2 hydrogel; [a] Higuchi, [b] Hixson-Crowell and [c] Korsmeyer-Peppas.

Table 1
Kinetics parameters for Famotidine release from Su-CNC/PNIPAm2 hydrogel.

Samples	Higuchi		Hixson-Crowell		Korsmeyer-Peppas		
	R ²	K _H × 10 ⁻² (h ^{-1/2})	R ²	K _{HC} × 10 ⁻² (h ^{-1/3})	R ²	n	K _{KP} × 10 ⁻² (h ⁻ⁿ)
pH 2	0.94	33.60	0.95	96.60	0.99	0.01	46.80
pH 10	0.96	10.20	0.98	25.80	0.99	0.01	13.80

4. Conclusion

Herein, this work presented a novel way to obtain multi responsive hydrogel through free chain polymerization reaction of the modified CNC with succinic acid in the presence of NIPAm. Different DS of Su-CNC were used to investigate their effects on the ESR % of the prepared hydrogel. From results, the SEM showed the pore size of the prepared hydrogels in range of 5–20 μm. However, the hydrogel Su-CNC/PNIPAm 2, prepared at DS of Su-CNC 1.8 represented the highest swelling as well as a remarkable high response to the pH trigger. The latter hybrid hydrogel manifested the highest thermal stability with lowest weight loss as per TGA. At this DS, the hydrogel formed exhibited pH and temperature responsiveness that swelled in alkaline medium and de-swelled in acidic medium. While at low temperature, the hydrogel swelled, but by increasing the temperature above 35 °C, the ESR of the prepared hydrogel decreased gradually by further elevation in temperature. The selected hydrogel was used as a template for In vitro Famotidine drug release study by exploitation of their pH and temperature sensitive properties. Results revealed that the drug release % enhances by altering the pH from 10 to 2 at 25 °C over a period more than 6 h. Importantly, the obtained hybrid hydrogels based Su-CNC/ PNIPAm 2 can be advocated as a novel intelligent candidate for pH and temperature responsive drug release.

CRedit authorship contribution statement

Hossam E. Emam: Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **Tharwat I. Shaheen:** Conceptualization, Methodology, Investigation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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